

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

MLT

Section Id :	640653138833
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	15
Number of Questions to be attempted :	15
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653387233
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532125923 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING

TECHNIQUES (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536927842. ✓ Yes

6406536927843. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653387234

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406532125924 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

Consider a dataset for a binary classification problem with five features, each of which is binary. If we go for a generative model that satisfies the class-conditional independence assumption, then how many free parameters do we need to learn?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

Question Number : 3 Question Id : 6406532125931 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

Consider a regression problem in which we use a neural network with the following architecture:

- Input layer has ¹⁰ neurons
- Two hidden layers, each with ³⁰ neurons
- Output layer with one neuron

Find the total number of parameters in the network. Ignore the biases.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1230

Question Number : 4 Question Id : 6406532125932 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

In the context of Bayesian estimation, consider a dataset D having twenty points, fifteen of which are 1 and the rest are 0. We assume that a Bernoulli distribution with parameter p generates the data. The prior for p is assumed to be $\text{Beta}(3, 2)$. Find a point-estimate for the parameter p using the expected value of the posterior. Enter your answer correct to two decimal places.

Note: Recall that the expected value of a random variable following $\text{Beta}(\alpha, \beta)$ is given by $\frac{\alpha}{\alpha + \beta}$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.71 to 0.73

Sub-Section Number :

3

Sub-Section Id :

640653387235

Question Shuffling Allowed :

Yes

Question Number : 5 Question Id : 6406532125925 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

In the context of bagging, consider a dataset of size n . A bag is a set of n points that is obtained by sampling uniformly at random from the dataset with replacement. Which of the following represents the probability that a data-point (x_i, y_i) is not a part of the bag?

Options :

6406536927845. ✖ $\frac{1}{n}$

6406536927846. ✖ $1 - \frac{1}{n}$

6406536927847. ✔ $\left(1 - \frac{1}{n}\right)^n$

6406536927848. ✖ $1 - \left(1 - \frac{1}{n}\right)^n$

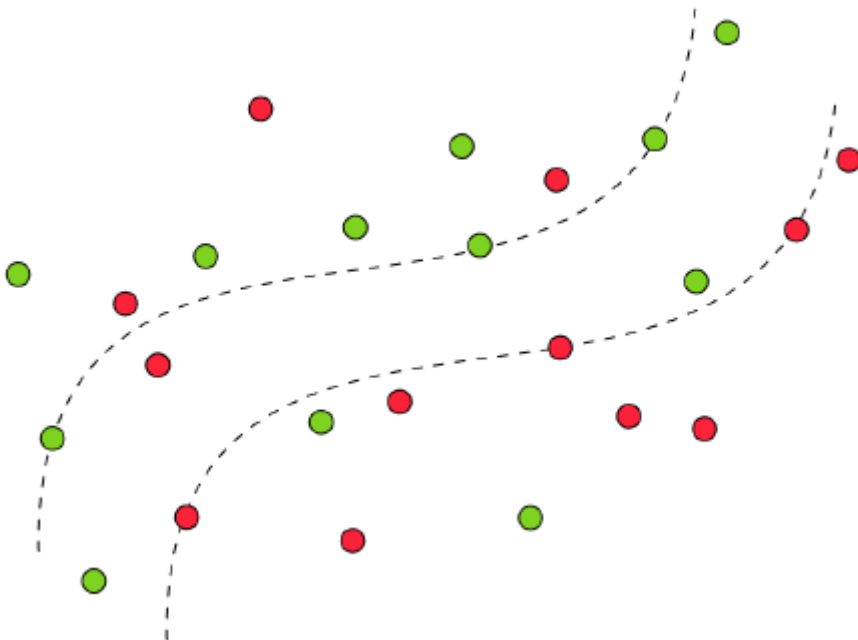
Question Number : 6 Question Id : 6406532125935 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a dataset for a binary classification problem with features in $\mathbb{R}^2$ as shown in the figure.

The red points belong to class -1 while the green points belong to class $+1$. Which of the following classifiers is the most appropriate choice?



Options :

6406536927866. ✖ Hard-margin, Linear SVM

6406536927867. ✖ Hard-margin, Kernel SVM, with a suitable kernel

6406536927868. ✖ Soft-margin, Linear SVM

6406536927869. ✔ Soft-margin, Kernel SVM, with a suitable kernel

Sub-Section Number :

Sub-Section Id :

640653387236

Question Shuffling Allowed :

No

Question Id : 6406532125926 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (7 to 8)

Question Label : Comprehension

Consider a soft-margin SVM that has been trained on a dataset with $C = 0.5$, where C is the hyperparameter that controls the tradeoff between the margin and the hinge loss. Let w^* be the optimal weight vector. Consider a data-point (x, y) in the dataset for which $(w^{*T} x)y = 0.7$. ξ^* is the slack corresponding to this point and α^* is the multiplier corresponding to the primal constraint for this point.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 7 Question Id : 6406532125927 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

Find the value of ξ^* . Enter your answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.3

Question Number : 8 Question Id : 6406532125928 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

Find the value of α^* . Enter your answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Sub-Section Number :

5

Sub-Section Id :

640653387237

Question Shuffling Allowed :

Yes

Question Number : 9 Question Id : 6406532125929 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a polynomial kernel, $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ of degree p . Let $x, y \in \mathbb{R}^d$. Which of the following is/are true?

Options :

6406536927851. ✓ If p is an even number, then $k(x, y) \geq 0$ for all $x, y \in \mathbb{R}^d$.

6406536927852. ✗ If x and y are orthogonal, then $k(x, y) = 0$.

6406536927853. ✗ $k(x, y)$ can never be zero.

6406536927854. ✓ $k(x, y) = k(y, x)$ for all $x, y \in \mathbb{R}^d$.

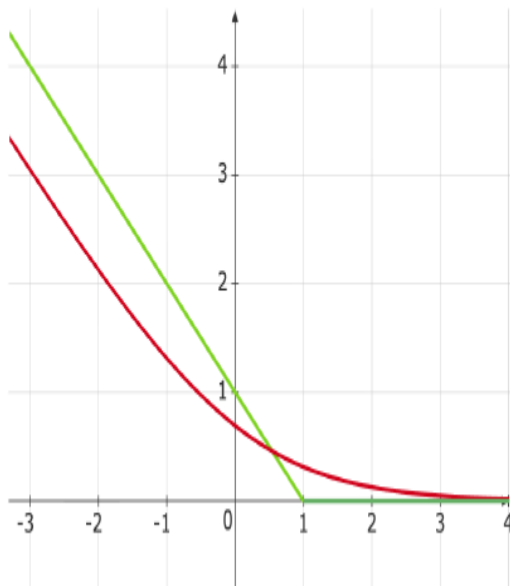
Question Number : 10 Question Id : 6406532125930 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Two loss functions are plotted in this graph. What are the two?

Note: On the horizontal-axis, we have the quantity $u = (w^T x)y$. The vertical axis is the loss expressed as a function of u .



Options :

6406536927855. ✓ Logistic loss

6406536927856. ✓ Hinge loss

6406536927857. ✗ Squared loss

6406536927858. ✗ 0-1 loss

Question Number : 11 Question Id : 6406532125934 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let C be the covariance matrix of a centered dataset with n data-points in $\mathbb{R}^d$, where X is the data-matrix of shape $d \times n$. Let w_i be the i^{th} principal component. Which of the following is/are true?

Options :

6406536927862. ✓ $w_i^T C w_i$ is the variance of the dataset along w_i

6406536927863. ✗ $C = \frac{1}{n} X^T X$

6406536927864. ✓ $w_i^T w_j = 0$, for $i \neq j$

6406536927865. ✗ $w_i^T C w_i > 0$ for each i satisfying $1 \leq i \leq n$

Sub-Section Number :

6

Sub-Section Id :

640653387238

Question Shuffling Allowed :

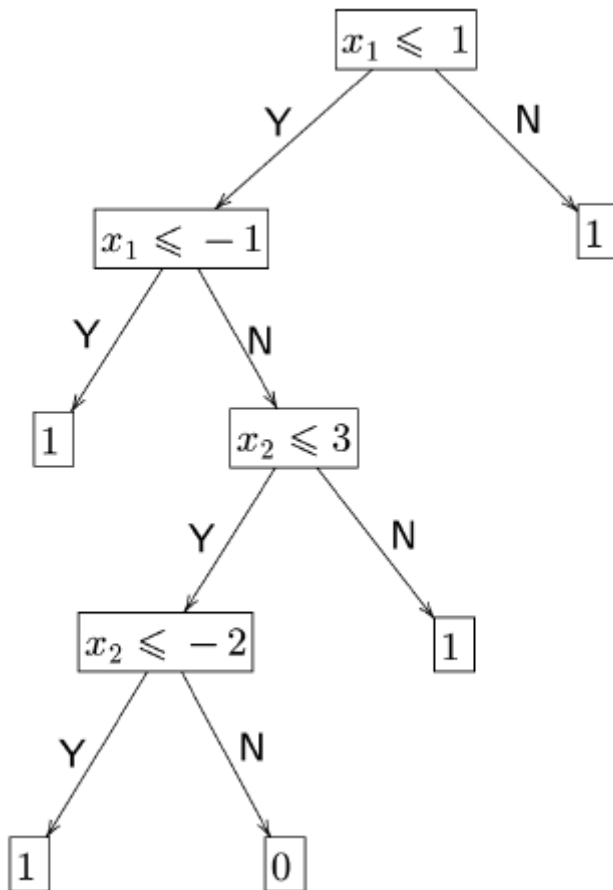
Yes

Question Number : 12 Question Id : 6406532125933 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 4

Question Label : Short Answer Question

Consider the following decision tree for a binary classification problem with features in $\mathbb{R}^2$ and labels in $\{1, 0\}$.



If S denotes the set of all points in $\mathbb{R}^2$ for which the decision tree outputs 0 as the predicted label, find the area corresponding to the region S . Enter the nearest integer as your answer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Number : 13 Question Id : 6406532125940 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 4

Question Label : Short Answer Question

Consider a logistic regression classifier trained on a dataset for a binary classification problem with features in $\mathbb{R}^4$ and labels in $\{0, 1\}$. The weight vector learnt by the classifier is $w \in \mathbb{R}^4$. Given a test data-point $x_{\text{test}} \in \mathbb{R}^4$, the probability $P(y = 1 \mid x_{\text{test}}) = \frac{e^2}{1 + e^2}$, where e is the base of the natural logarithm. Find the value of $w^T x_{\text{test}}$ and enter the nearest integer as your answer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Sub-Section Number :

7

Sub-Section Id :

640653387239

Question Shuffling Allowed :

No

Question Id : 6406532125936 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (14 to 16)

Question Label : Comprehension

Consider a linearly separable dataset, D , for a binary classification problem with features in $\mathbb{R}^2$ and labels in $\{1, -1\}$ that has a positive margin. D has ten data-points out of which five of them belong to the class $+1$. Assume that all ten data-points occupy distinct locations in $\mathbb{R}^2$.

Let $w \in \mathbb{R}^2$ be the optimal weight vector output by a hard-margin SVM. Consider three subsets of D as given below:

$$D_1 = \{(x_i, y_i) : (w^T x_i) y_i < 1, (x_i, y_i) \in D\}$$

$$D_2 = \{(x_i, y_i) : (w^T x_i) y_i > 1, (x_i, y_i) \in D\}$$

$$D_3 = \{(x_i, y_i) : (w^T x_i) y_i = 1, (x_i, y_i) \in D\}$$

$|D_i|$ denotes the cardinality of the set. Note that $|D| = 10$. Answer the given sub-questions.

Sub questions

Question Number : 14 Question Id : 6406532125937 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

What is $|D_1|$? If you cannot compute this value with the given information, enter -1 as the answer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 15 Question Id : 6406532125938 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 2

Question Label : Short Answer Question

What is $|D_2| + |D_3|$? If you cannot compute this value with the given information, enter -1 as the answer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Number : 16 Question Id : 6406532125939 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true?

Options :

6406536927872. ✖ $|D_3| = 2$

6406536927873. ✔ $|D_3| \geq 1$

6406536927874. ✖ $|D_2| \geq 1$

6406536927875. ✔ $|D_2| \geq 0$

Sub-Section Number :

8

Sub-Section Id :

640653387240

Question Shuffling Allowed :

Yes

Question Number : 17 Question Id : 6406532125941 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider a dataset with two data-points for a binary classification problem with features in $\mathbb{R}^2$. $(1, 0)$ is a data-point from the positive class while $(-1, 0)$ is from the negative class. We now consider three linear classifiers: perceptron, logistic regression and SVM.

Use gradient ascent to train the logistic regression model with $\eta = 1$. For both perceptron and logistic regression, run exactly one iteration of their respective algorithms starting with $w^{(0)} = (0, 0)$ in each case. That is, you should stop after you get $w^{(1)}$. Train the SVM in the usual way.

Use 0 as the negative class label for logistic regression and -1 for SVM and perceptron. Use the following rule for determining the predicted labels for the three classifiers:

$$\begin{array}{c|c} \text{SVM and Perceptron} & \text{Logistic Regression} \\ \hline \hat{y} = \begin{cases} 1, & w^T x \geq 0 \\ -1, & w^T x < 0 \end{cases} & \hat{y} = \begin{cases} 1, & w^T x \geq 0 \\ 0, & w^T x < 0 \end{cases} \end{array}$$

If $w^* \in \mathbb{R}^2$ is the weight vector with the largest margin, which of the following is true?

Options :

6406536927877. ✓ All three classifiers output w^*

6406536927878. ✗ Only SVM outputs w^*

6406536927879. ✗ Only SVM and perceptron output w^*

6406536927880. ✗ Only perceptron and logistic regression output w^*

Sub-Section Number :

9

Sub-Section Id :

640653387241

Question Shuffling Allowed :

Yes

Question Number : 18 Question Id : 6406532125942 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a dataset consisting of n datapoints $x_1, x_2, \dots, x_n$, where each $x_i \in \mathbb{R}^4$. Let the four features be denoted by f_1, f_2, f_3 , and f_4 . Suppose the features satisfy the following linear relationship:

$$f_3 = 2f_1 + 3f_4$$

That is, one of the features is an exact linear combination of others. A linear regression model is trained to predict a target variable y using these features. Consider the following statements: Consider the following statements: Which of the following is (are) true?

Options :

6406536927881. ✓ Linear regression without regularization will achieve training MSE less than or equal to that of ridge regression.

6406536927882. ✓ The solution for linear regression without regularization may not be unique.

6406536927883. ✗ Ridge regression (L2 regularization) will always set at least one coefficient exactly to zero.

6406536927884. ✓ Lasso regression (L1 regularization) can force some feature weights to become exactly zero.